

CodeTrain 16-Week AI Curriculum Strategy

The Blueprint for Scaling Python/AI Training Across 65+ Students

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1 Executive Summary

This document is the **Root Blueprint** for delivering the CodeTrain AI curriculum across **16 weeks** to **65+ students** in multiple cohorts (weekday, weekend, online). It transforms the original 2-week Training of Trainers (ToT) materials into a structured, paced, classroom-ready program with clear learning objectives, critical gates, and assessment milestones.

2 The 4-Phase Structure

The curriculum is organized into 4 distinct phases, each 4 weeks long, culminating in a milestone demo.

2.1 Phase 1: Foundations (Weeks 1-4)

Learning Goal: Students master Python syntax, data structures, and control flow as JavaScript developers transitioning to Python.

Key Principle: “You already know logic. This is just new syntax.”

Week	Content	Focus
1	Welcome + Environment Setup	Installation, uv, VS Code
2	Version Control Fundamentals	Git/GitHub for AI portfolios
3	Python Basics (Variables, Data Structures)	Lists, dicts, tuples, sets; JS-to-Python comparisons
4	Functions, Scope, Pythonic Patterns	Functions, comprehensions, modules; OOP intro

Module Mapping:

- 00-welcome: Welcome Guide (PDF)
- 01-setup: Setting Up Your Environment (PDF)
- 02-version-control: Git & GitHub Guide (PDF)
- 03-python-basics: Python Basics Parts 1-3 (PDFs) + Presentation Slides

Critical Gate 1 (End of Week 4): Python Mastery

- • Students write list comprehensions naturally (without looking them up)
- • Function scope (LEGB) is automatic, not confusing
- • Code reads as Pythonic, not JavaScript-in-Python
- • Mini Demo Day 1: Simple Logic Script (CLI tool, e.g., voting system, expense tracker)

Learning Objectives (Phase 1):

1. Understand why Python dominates AI/ML (ecosystem, speed of prototyping)
2. Install Python 3.12, uv, and VS Code; create isolated environments
3. Use Git/GitHub for version control and portfolio building
4. Write Pythonic code: f-strings, comprehensions, functions, type hints
5. Understand variable scope (local, enclosing, global, built-in)
6. Master basic OOP: classes, inheritance, error handling

2.2 Phase 2: Analytics (Weeks 5-8)

Learning Goal: Students build data pipelines, clean messy data, and extract insights using Pandas and visualization.

Key Principle: “Data science is 80% cleaning, 20% modeling. Master the cleaning first.”

Week	Content	Focus
5	Pandas Fundamentals	DataFrames, Series, loading data, indexing
6	Data Visualization & Transformation	Matplotlib, Seaborn, data types, chaining
7	Tidy Data & Groupby	Reshaping, groupby, aggregation, merging
8		

Week	Content	Focus
	Mini Demo Day 2 Prep & Presentation	EDA challenge, report writing, presentation skills

Module Mapping:

- 04-data-analysis: Pandas Parts 1-3 (PDFs) + Presentation Slides

Critical Gate 2 (End of Week 8): Pandas Fluency

- • `.loc[]` vs `.iloc[]` is automatic (no syntax lookup)
- • Groupby and aggregation are intuitive
- • Data questions are answered without external resources
- • Mini Demo Day 2: Exploratory Data Analysis Report (notebook + PDF + 5-slide summary)

Learning Objectives (Phase 2):

1. Load data from CSV, JSON, Excel; explore with `.head()`, `.describe()`, `.info()`
2. Handle missing values, duplicates, and data quality issues
3. Select, filter, and manipulate DataFrames using `.loc[]`, `.iloc[]`, boolean indexing
4. Create publication-ready visualizations: line plots, scatter, histograms, heatmaps
5. Write data pipelines using chaining: `df.filter().sort().groupby().agg()`
6. Reshape data with `.melt()`, `.pivot_table()`, and normalization

2.3 Phase 3: Intelligence (Weeks 9-12)

Learning Goal: Students train, evaluate, and deploy machine learning models on real-world datasets.

Key Principle: “ML is applied statistics. Understand the data before optimizing the model.”

Week	Content	Focus
9	Machine Learning Fundamentals	Train/test split, loss, metrics, linear regression
10	Classification & Decision Boundaries	Logistic regression, decision trees, confusion matrices
11		

Week	Content	Focus
	Ensemble Methods & Hyperparameter Tuning	Random Forest, XGBoost, grid search, feature importance
12	Mini Demo Day 3 Prep & Presentation	Model selection, evaluation, interpretation

Module Mapping:

- 05-machine-learning: ML Parts 1-3 (PDFs) + Presentation Slides + Exercise PDFs

Critical Gate 3 (End of Week 12): Model Building Competency

- • Train/test split is automatic (no leakage, no confusion)
- • Model evaluation metrics make sense (precision vs recall, F1)
- • Hyperparameter tuning is methodical (grid search, not guessing)
- • Mini Demo Day 3: Predictive Model Build (trained model, evaluation metrics, code notebook)

Learning Objectives (Phase 3):

1. Understand the ML workflow: problem framing, feature engineering, train/test split, evaluation
2. Implement linear regression from scratch and using sklearn
3. Train classifiers (logistic regression, decision trees) and interpret predictions
4. Evaluate models using confusion matrices, precision, recall, F1, ROC-AUC
5. Combat overfitting using regularization, cross-validation, and ensemble methods
6. Use Random Forest and XGBoost for tabular data; tune hyperparameters with GridSearchCV
7. Extract feature importance and explain model predictions

2.4 Phase 4: Advanced AI (Weeks 13-16)

Learning Goal: Students build end-to-end deep learning solutions and understand agentic AI paradigms.

Key Principle: “Deep learning unlocks unstructured data: images, text, sequences. Agents unlock reasoning.”

Week	Content	Focus
13	Deep Learning Fundamentals	

Week	Content	Focus
		Neural networks, activation functions, backpropagation, PyTorch
14	CNNs & Image Classification	Convolutions, pooling, transfer learning, Google Colab
15	Advanced AI: Agents & Orchestration	LLMs, RAG, agentic workflows, CodeTrain framework
16	Final Demo Day & Capstone	Team presentations, code review, deployment walkthrough

Module Mapping:

- 06-deep-learning: DL Parts 1-3 (PDFs) + Presentation Slides + Exercise PDFs
- 07-agentic-ai: Agentic AI Course Materials (PDFs) + Presentation Slides

Final Demo Day (End of Week 16): Capstone Project

- • Working AI application (image classifier, NLP agent, or predictive model)
- • Real dataset, real model, real inference
- • GitHub repository with documented code
- • 15-min team presentation + 10-min Q&A
- • Portfolio-ready for employers

Learning Objectives (Phase 4):

1. Build and train neural networks in PyTorch: layers, activation functions, loss functions, optimizers
2. Understand backpropagation conceptually and debug training loops
3. Implement CNNs for image classification; understand pooling and dropout
4. Use transfer learning: load pre-trained weights, fine-tune on new data
5. Understand LLMs, Retrieval Augmented Generation (RAG), and orchestration frameworks
6. Build AI agents: decision loops, tool use, workflow orchestration
7. Deploy models to production: model export, API serving, inference at scale

3 Critical Learning Gates

These gates act as competency checkpoints. If a student doesn't pass a gate, they cannot meaningfully proceed to the next phase.

3.1 Gate 1: Python Mastery (End of Week 4)

Observable Behavior:

- Student writes a 20-line function with list comprehensions, no syntax lookup, no StackOverflow visits
- Student uses `.format()` naturally; understands when to use `*args` vs `**kwargs`
- Student's code contains type hints and docstrings (Google style)
- Error messages don't confuse them; they use tracebacks to debug

If a student struggles: Assign supplementary Python Exercises (PDF). Do NOT proceed to Pandas until they pass.

Assessment: Mini Demo Day 1 (Python Logic Script) — must score 18+ out of 25 points.

3.2 Gate 2: Pandas Fluency (End of Week 8)

Observable Behavior:

- Student writes `.loc[mask, column]` without hesitation
- Student groups data, aggregates, and reshapes without external help
- Student answers "How many transactions per user?" by writing `df.groupby('user').size()` automatically
- Student creates 3-panel visualizations without copy-pasting code

If a student struggles: Assign supplementary Pandas Exercises (PDF). Pair with strong students for study groups.

Assessment: Mini Demo Day 2 (EDA Report) — must score 18+ out of 25 points.

3.3 Gate 3: Model Building Competency (End of Week 12)

Observable Behavior:

- Student trains/tests without leaking (they use `.fit()` on train only, never on test)
- Student understands precision vs recall and chooses metrics for their use case
- Student uses GridSearchCV to tune hyperparameters (not random guessing)
- Student can explain feature importance and model limitations

If a student struggles: Assign ML Exercises (PDF). Offer 1:1 mentorship on model interpretation.

Assessment: Mini Demo Day 3 (Predictive Model) — must score 18+ out of 25 points.

4 Content Mapping: All 19 Notebooks Across 4 Phases

Phase	Week(s)	Notebook	Module	Hours
Phase 1	1	python-vs-other-languages	00-welcome	1
Phase 1	1	setting_up	01-setup	2
Phase 1	2	Git_Github	02-version-control	3
Phase 1	3-4	intro_python1	03-python-basics	4
Phase 1	3-4	intro_python2	03-python-basics	5
Phase 1	4	intro_python3	03-python-basics	5
Phase 1	4	python_basics_presentation	03-python-basics	1 (recap)
Phase 2	5-6	intro_pandas1	04-data-analysis	5
Phase 2	6-7	intro_pandas2	04-data-analysis	5
Phase 2	7-8	intro_pandas3	04-data-analysis	5
Phase 2	8	data_analysis_presentation	04-data-analysis	1 (recap)
Phase 3	9-10	intro_ML1	05-machine-learning	5

Phase	Week(s)	Notebook	Module	Hours
Phase 3	10-11	intro_ML2	05-machine-learning	5
Phase 3	11-12	intro_ML3	05-machine-learning	5
Phase 3	12	machine_learning_presentation	05-machine-learning	1 (recap)
Phase 4	13-14	intro_DL1	06-deep-learning	6
Phase 4	14	intro_DL2	06-deep-learning	6
Phase 4	15	intro_DL3	06-deep-learning	6
Phase 4	15	agentic-ai (all 5 files)	07-agentic-ai	8

Total: ~95 contact/study hours across 16 weeks (~6 hrs/week)

5 How to Use This Curriculum

5.1 For Instructors (Richard & Joshua)

1. **Pre-Week Prep:** Review the week's content from the delivery schedule
2. **Live Sessions:** Follow the session plans; use the slide presentations for motivation, the notebooks for hands-on coding
3. **Exercise Time:** Direct students to the supplementary exercises (Python, Pandas, ML Exercise PDFs)
4. **Assessment:** Use the milestone rubrics to grade Demo Days fairly and transparently
5. **Office Hours:** Address common struggles (see Trainer Guide for typical pain points)

5.2 For Students

1. **Before Class:** Read the learning objectives for the week
2. **During Class:** Follow along with the instructor; take notes on the “Why” sections
3. **After Class:** Complete the exercises and self-check against solutions
4. **Before Milestone:** Prep your Demo Day submission using the rubric as your checklist

5.3 For Cohort Managers

1. **Weekday Cohort:** 3 × 1.5hr sessions per week (Mon, Wed, Fri afternoon or Tue, Thu morning + 1 more)
2. **Weekend Cohort:** 1 × 4hr session per week (Saturday morning) + optional Sunday 1hr office hours
3. **Online Cohort:** Async videos + 1 × 2hr live Q&A per week (Monday evening)

See the Delivery Schedule PDF for detailed session timing and content.

6 The Learning Philosophy

Why → How → Why Next

Every lesson follows this arc:

- **Why:** The historical problem solved by this concept (e.g., “Why do we need functions?”)
- **How:** The implementation and code examples
- **Why Next:** The evolution and what comes after (e.g., “Functions lead to modules; modules lead to classes”)

This bridges theory and practice. Students don’t just learn *what* to code—they understand *why* the concept exists.

7 Quality Assurance

7.1 Before Rolling Out:

- ☐ All 19 notebooks render without errors to PDFs
- ☐ All `eval: true` code cells in exercise notebooks run without exceptions
- ☐ Each phase has a well-defined milestone with clear submission format

- ☐ Trainer guide reviewed by Richard & Joshua for accuracy
- ☐ Delivery schedule cross-checked: all weeks 1-16 accounted for, all notebooks assigned

7.2 During Delivery:

- ☐ Weekly pulse checks: Is the cohort hitting learning gates?
- ☐ Feedback loops: Students struggle → adjust pacing for next cohort
- ☐ Capstone projects: Real dataset, real problem, real evaluation

8 Success Metrics

By the end of 16 weeks, 80%+ of students should be able to:

1. **Phase 1:** Write Pythonic code; understand scope and imports
2. **Phase 2:** Load, clean, visualize, and present data insights
3. **Phase 3:** Train, evaluate, and interpret machine learning models
4. **Phase 4:** Build and deploy end-to-end AI solutions

Final outcome: Portfolio-ready projects that students can show employers in Ghana and Europe.

9 Next Steps

1. Review the Delivery Schedule PDF for week-by-week session plans
2. Study the Milestone Rubrics PDF to understand assessment criteria
3. Read the Trainer Guide PDF for pedagogical tips and common struggles
4. Use supplementary exercise PDFs (Python, Pandas, ML, DL) to fill curriculum gaps
5. Schedule mentorship sessions with Richard & Joshua (2 days of the 7-day scope)

Created as part of the CodeTrain Flexible Remuneration extension (7-day scope). This document is the root blueprint for 16-week delivery to 65+ students across weekday, weekend, and online cohorts.